Dr. Ivan V. Arzhantsev
Associate Professor
Higher Algebra Department
Faculty of Mechanics
and Mathematics
Moscow State University
Leninskie Gory, GSP-1 Moscow
119991 Russia
arjantse@mccme.ru

March 5, 2009

Recommendation Letter

This is to recommend Alexander Perepechko for participation in Summer School "*Structures in Lie Representation Theory*".

I know Alexander since September 2004. He was the best student in my algebra classes for the first and later for the second year students at Moscow State University. During these classes we discussed some difficult problems, and many times Alexander found short and beautiful arguments that were really new and unexpected for me.

Since September 2006, Alexander is a student of Higher Algebra Department, and I am his scientific supervisor. Alexander's first research work was devoted to endomorphisms' monoids of finite dimensional algebras. Recently, N.L.Gordeev and V.L.Popov proved that any affine algebraic group can be realized as the automorphism group of a finite dimensional algebra (Annals of Math. 158 (2003), 1041-1065). Perepechko shows that any affine algebraic monoid can be realized as the endomorphisms' monoid of a finite dimensional algebra that contains zero as an isolated component. Following the scheme of Gordeev-Popov's work, Alexander introduces some new ideas and constructions that are specific for the case of monoids. Now this result is accepted for publication in the Proceedings of the American Mathematical Society.

Current Alexander's work is connected with classification of affine algebraic monoids with a given solvable group as the group of invertible elements and with the automorphism group of a commutative finite dimensional local algebra. I hope that some interesting results will be obtained soon.

Alexander is an active participant of our seminars. His talks are well-prepared and clearly presented, and it is a pleasure to work with Alexander in the framework of the seminars.

In my opinion, Alexander Perepechko is a gifted and hard-working young mathematician. I think that he will fully benefit from attending the summer school, and I strongly recommend Alexander for this event.